

MID EXAM

Course: Data structures (cs01)

Set: A | Time: 1 Hour | Max Marks: 41 | Date: 14-03-2026

Total Questions: 9 | Answer all questions.

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

SECTION A: MCQs

1. Which of the following statements accurately describes an implication of the software architecture definition provided in the text? [4 Marks]

- a) An architecture must explicitly capture all desired system properties to be considered complete.
- b) Every computing system with software necessarily has an architecture that is known and documented.
- c) Software architecture is primarily an abstraction that suppresses details of elements that do not affect their interaction with other elements.
- d) The definition of software architecture intrinsically evaluates whether the architecture is good or bad.

2. Based on the text, what is the fundamental difference between an "architectural pattern" and an "architecture"? [4 Marks]

- a) An architectural pattern describes elements and relations for a specific system, while an architecture is a set of constraints.
- b) An architectural pattern is a standard decomposition of a known problem, whereas an architecture is a mapping of functional requirements to a set of components.
- c) An architectural pattern defines a set or family of architectures, but is not an architecture itself.
- d) An architectural pattern focuses on runtime elements, while an architecture focuses on compile-time implementation units.

3. A software development team is designing a system and needs to represent elements that have a runtime presence, such as processes and data stores, along with their communication pathways. Which type of architectural view is most appropriate for this purpose according to the text? [4 Marks]

- a) Module View
- b) Allocation View
- c) Quality View
- d) Component-and-Connector (C&C) View

4. According to the text, which of the following is a direct benefit of designing for "testability" as a system quality attribute? [4 Marks]

- a) It ensures the system can run under different computing environments without modification.
- b) It helps in detecting failure modes and can significantly reduce the overall project testing cost.
- c) It guarantees that the system is resistant to unauthorized attempts at usage or behavior modification.
- d) It primarily concerns the system's response times and throughput under various loads.

SECTION A: SHORT ANSWERS

5. In the Software Architecture Business Cycle, besides technical skill, what three essential non-technical skills must architects possess to effectively engage with stakeholders and manage the architectural process? [5 Marks]

6. Explain the concept of a "View" in software architecture, emphasizing what it represents and what it explicitly excludes. **[5 Marks]**
7. Describe the relationship between an "architectural pattern" and a "tactic" as explained in the text. **[5 Marks]**
8. Identify and describe three distinct uses or applications of architectural reconstruction results, as outlined in the provided text. **[5 Marks]**
9. When refining a module using the Attribute-Driven Design (ADD) approach, what are the first three consecutive steps involved in this refinement process? **[5 Marks]**
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